

# Matthew Tonks's Visualization Guide

## ***Comprehensive Guide to Data Visualization Types***

This playbook demonstrates visualization techniques organized by plot type, following Edward Tufte's principles of graphical excellence and Stephen Few's perceptual accuracy guidelines. Each section emphasizes maximizing data-ink ratio, reducing chart junk, and leveraging human visual perception strengths.

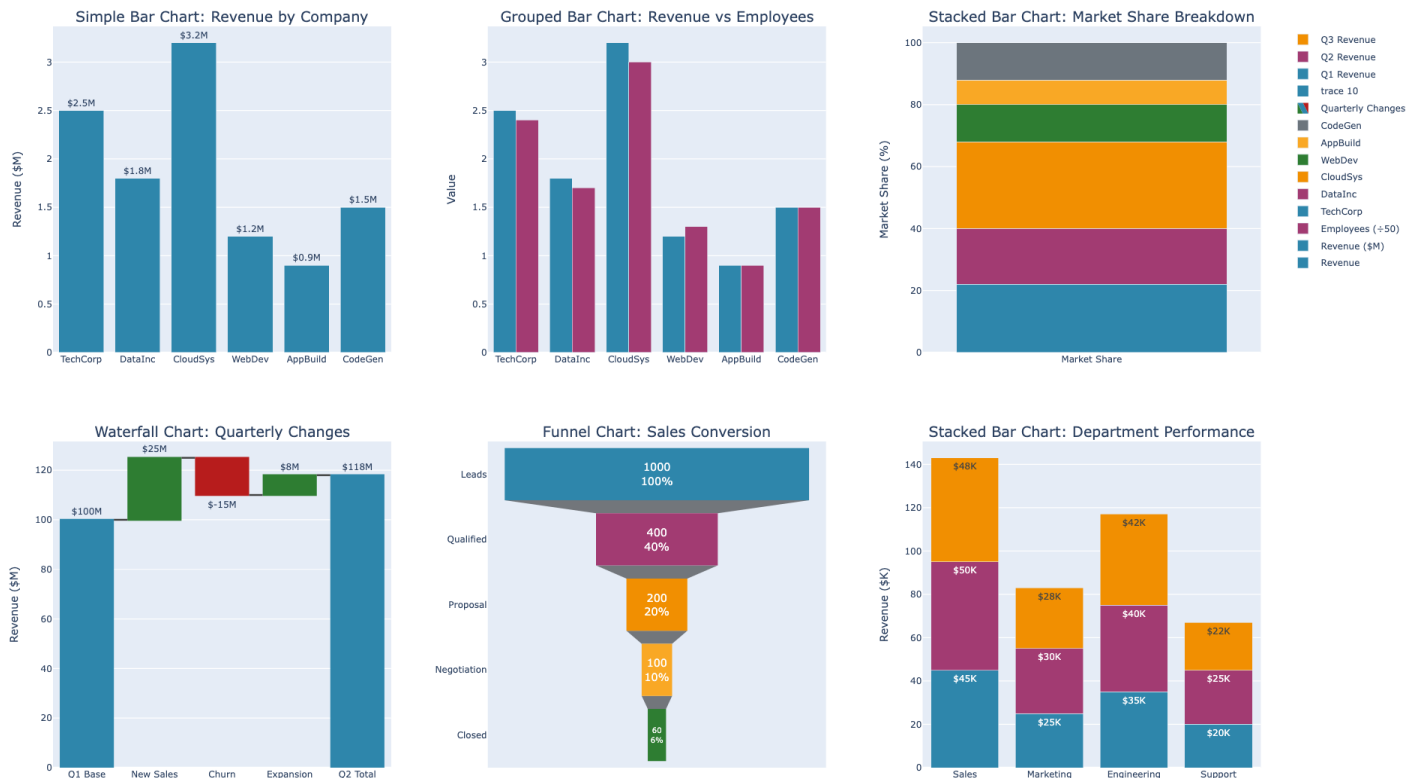
### ***Core Visualization Principles***

**Tufte's Graphical Excellence:** Show the data. Induce the viewer to think about the substance rather than methodology. Avoid distorting what the data has to say. Present many numbers in a small space. Make large data sets coherent. Encourage comparison of different pieces of data.

**Few's Perceptual Guidelines:** Leverage pre-attentive visual processing. Use position over length, length over angle, angle over area. Design for the colorblind. Reduce cognitive load through direct labeling and intuitive layouts.

# Categorical Plots

Categorical Data Visualization Types



## Plot Type Analysis:

**Purpose:** Compare discrete categories and their distributions.

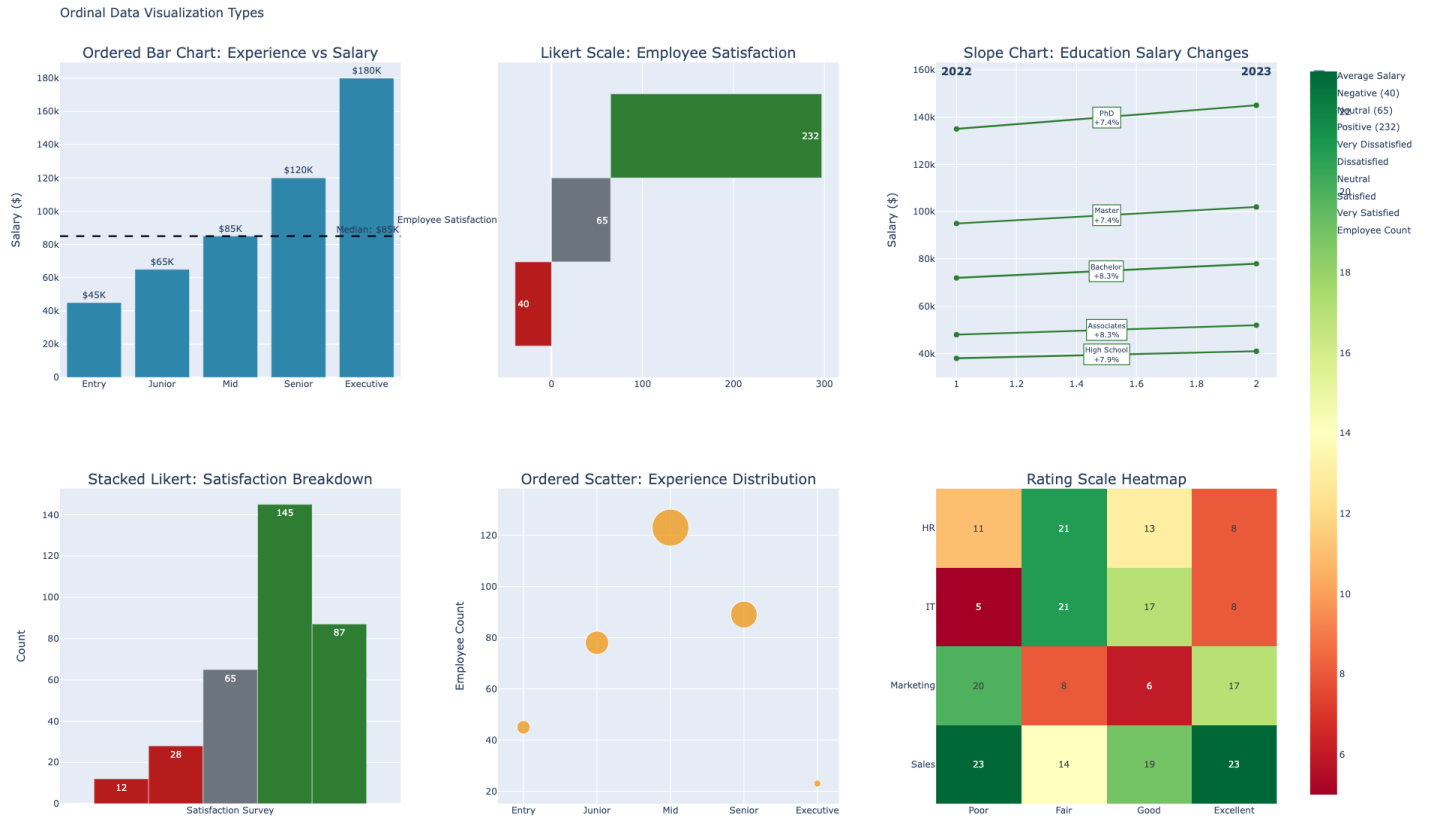
**Tufte's Data-Ink Ratio:** Maximize information content by removing chart junk. Use direct labeling instead of legends when possible. Eliminate unnecessary grid lines and decorative elements.

### Few's Simplicity Principles:

- Waterfall charts excel at showing cumulative changes - more informative than stacked bars
- Funnel charts clearly communicate conversion processes
- Sunburst charts reveal hierarchical patterns impossible to see in flat bar charts
- Remove pie charts - human visual system poorly estimates angles and areas

**Best Practices:** Sort categories by value (not alphabetically). Use consistent color schemes. Direct label values to reduce cognitive load. Start bars at zero baseline to avoid misleading comparisons.

# Ordinal Plots



## Plot Type Analysis:

**Purpose:** Visualize ordered categories, rankings, and scales with inherent sequence.

**Tufte's Sparklines Concept:** Small multiples and slope charts show change more effectively than complex multi-series plots. Remove chart borders and excess annotation.

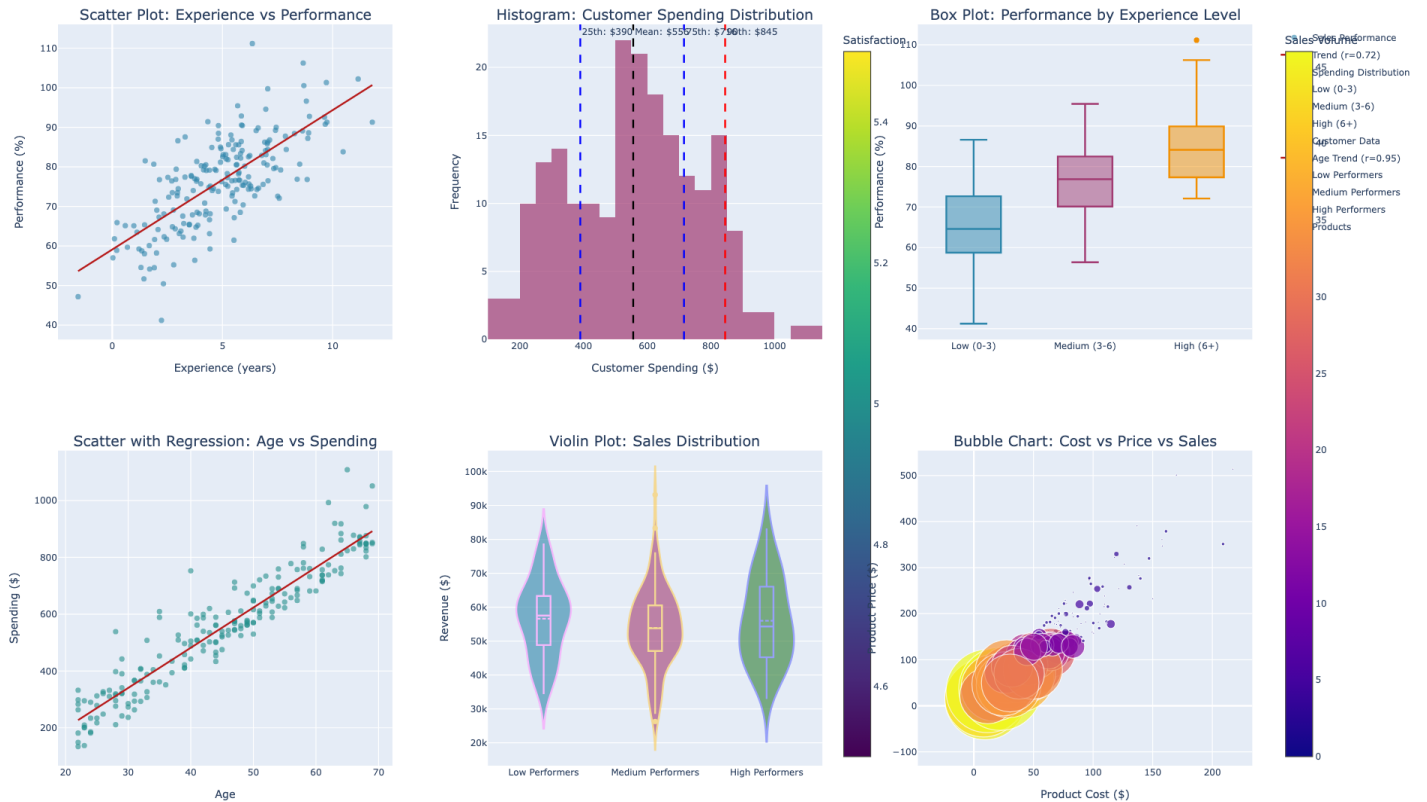
### Few's Perceptual Guidelines:

- Slope charts reveal changes better than line charts for discrete time points
- Heatmaps encode magnitude through color intensity - use sequential color schemes
- Dot plots often superior to bar charts for ordered categories
- Violin plots show distribution shape that box plots cannot capture

**Best Practices:** Maintain natural ordering. Use position over color for primary comparisons. Include reference lines for context (medians, targets). Gray out less important data to highlight key insights.

# Quantitative Plots

Quantitative Data Visualization Types



## Plot Type Analysis:

**Purpose:** Explore relationships, distributions, and patterns in continuous numerical data.

**Tufte's Analytical Focus:** Show statistical relationships clearly. Use small multiples to compare distributions across groups. Minimize non-data ink - let the data patterns dominate.

### Few's Accuracy Principles:

- Scatter plots reveal correlations that tables cannot show
- Box plots compress distribution information efficiently
- Histograms show shape - normal, skewed, bimodal distributions
- Avoid 3D effects and unnecessary depth that distort perception

**Best Practices:** Include trend lines for relationships. Mark outliers clearly. Use logarithmic scales when data spans orders of magnitude. Add rug plots to show actual data points beneath density curves.

# Temporal Plots

Temporal Data Visualization Types



## Plot Type Analysis:

**Purpose:** Reveal patterns, trends, cycles, and anomalies in time-ordered data.

**Tufte's Time-Series Excellence:** Time moves left to right - never violate this convention. Remove unnecessary time axis labels. Use horizon charts for small multiples with many series.

## Few's Temporal Clarity:

- Moving averages smooth noise to reveal underlying trends
- Multiple time series require careful color differentiation
- Seasonal patterns need consistent baseline years for comparison
- Event markers provide crucial business context

**Best Practices:** Start time axis at meaningful baseline. Use aspect ratios that don't exaggerate volatility. Annotate significant events directly on timeline. Consider cycle charts for seasonal data.

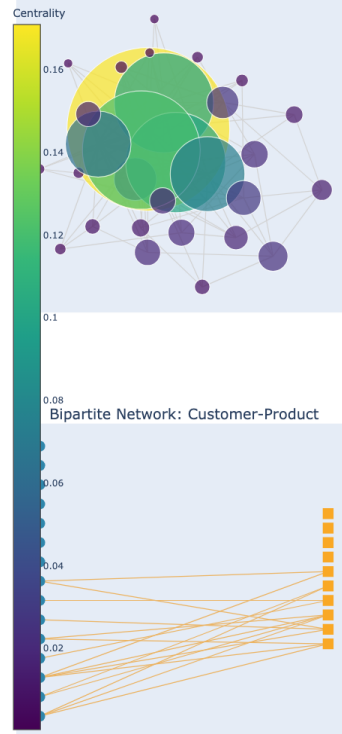
# Network Plots

Network/Relational Data Visualization Types

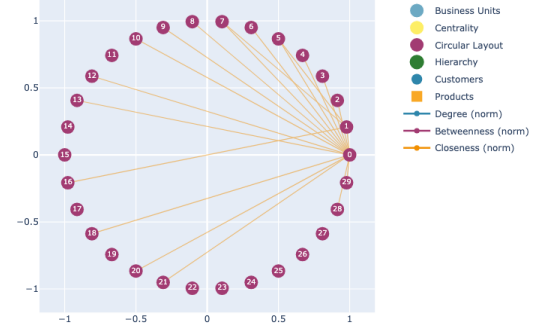
Node-Link Network: Business Connections



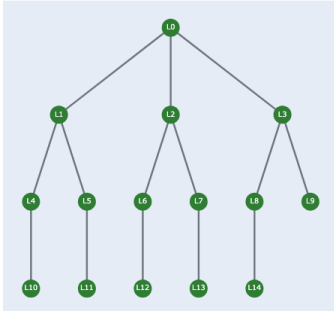
Centrality Analysis: Key Players



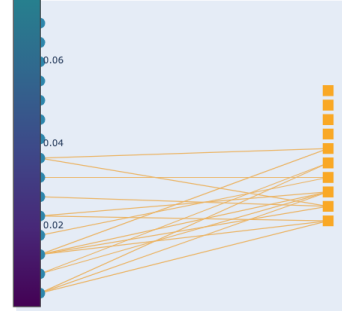
Chord Diagram Style: Relationships



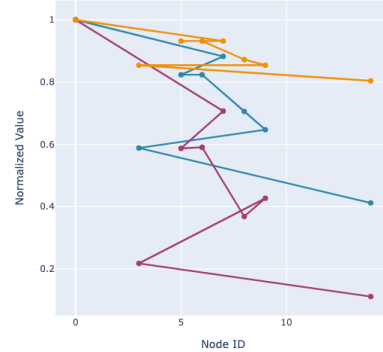
Hierarchical Network: Organizational



Bipartite Network: Customer-Product



Network Metrics Comparison



## Plot Type Analysis:

**Purpose:** Visualize relationships, connections, and structures between entities.

**Tufte's Network Principles:** Minimize edge crossings. Use node size to encode importance. Keep layouts consistent for comparison. Remove isolated nodes unless specifically relevant.

## Few's Relationship Clarity:

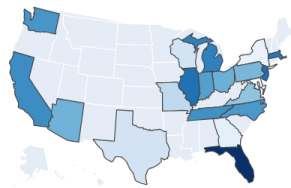
- Force-directed layouts reveal natural clustering
- Circular layouts work well for hierarchical relationships
- Bipartite networks separate entity types clearly
- Matrix representations often clearer than node-link for dense networks

**Best Practices:** Color-code node types consistently. Use edge thickness for relationship strength. Include centrality measures for business relevance. Consider matrix view for complex networks with many connections.

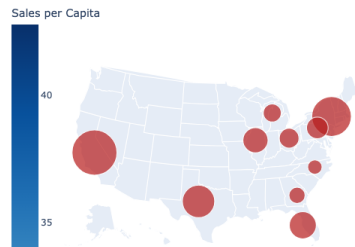
# Spatial Plots

Spatial Data Visualization Types

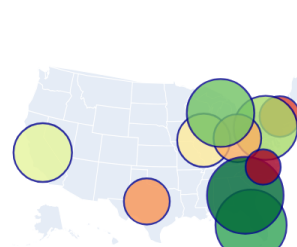
Choropleth Map: Sales per Capita



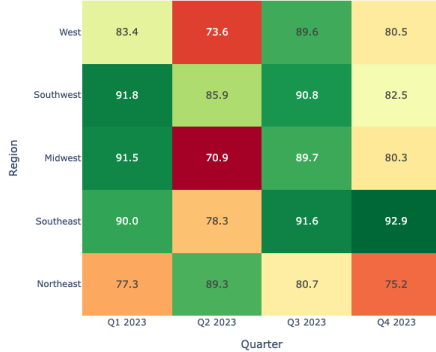
Scatter Geo: Store Locations



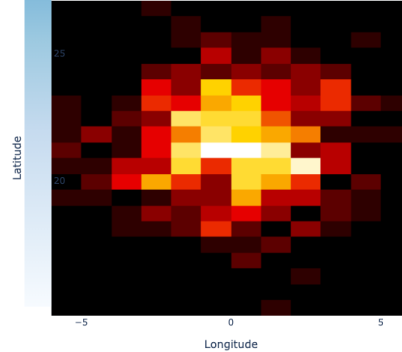
Bubble Map: Market Penetration



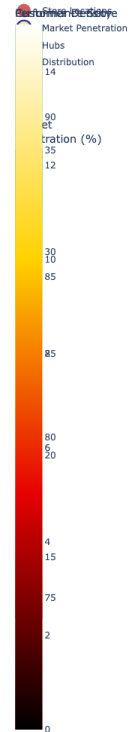
Heatmap: Regional Performance



Density Map Style: Customer Distribution



Geographic Network: Supply Chain



## Plot Type Analysis:

**Purpose:** Analyze geographic patterns, regional differences, and location-based business metrics.

**Tufte's Geographic Accuracy:** Maintain geographic proportions. Use choropleth maps for normalized data only. Avoid misleading area distortions. Include per-capita adjustments for population-based metrics.

### Few's Spatial Guidelines:

- Bubble maps better than choropleth for absolute values
- Density maps reveal patterns invisible in point data
- Supply chain networks show business flow geography
- Color schemes must be intuitive (red=bad, green=good)

**Best Practices:** Normalize by population/area when appropriate. Use consistent geographic projections. Include reference points (major cities). Design for colorblind accessibility with diverging palettes.